UtilOS® Firmware 5.4.0–5.4.3 Release Notes

UtilOS® Firmware 5.4.0–5.4.3 Release Notes
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REV 005
5.4

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New in This Document

Revision	Date	Description
REV 005	February 2024	Added 5.4.3 release notes
REV 004	January 2024	Corrected description in 5.4.2 for 2982237 in Resolved Issues on page 14.
REV 003	December 2023	Added 5.4.2 release notes
REV 002	August 2022	Updated Images table to move LCS from Zigbee to non-Zigbee for NIC 4xx.
REV 001	May 2022	Updated Support for Large Failover Groups in an AP-based DA network on page 24
REV 000	April 2022	First date of publication.

Firmware 5.4.3 Release Notes

Release Date: February 1, 2024

This release note describes resolved issues in the UtilOS firmware 5.4.3.

Firmware Images

These images can be obtained from https://products.itron.com.

Table 1 UtilOS firmware 5.4.3 images

Image	5.4.3 Device Version	Supported Devices	Supported Battery Powered Devices	NIC Version
apr15_arm.nic.image.prod.prod_sig.05.04.5003.03	5.4.5003	AP 1.5 Relay 1.5		NIC 2xx
slic_arm.nic.image.prod.prod_sig.05.04.4003.03	5.4.4003	eBridge 2.0		NIC 2xx
slic_akio.nic.image.prod.prod_sig.05.04.4023.03	5.4.4023	Sentient Energy MM2/MM3 FCI		NIC 3xx/4xx
aprel_akio.nic.image.prod.prod_sig.05.04.6023.03	5.4.6023	AP 1.5i Relay 1.5i Bridge 4.0 AP 4.5 Relay 4.5 MicroAP™ for Streetlight cabinet with EDMI 10E		NIC 4xx
aprel_akio.nic_z.image.prod.prod_sig.05.04.6033.03	5.4.6033	MicroAP (Zigbee)	IMU 500S-INS	NIC 4xx

Table 1 UtilOS firmware 5.4.3 images (continued)

Image	5.4.3 Device Version	Supported Devices	Supported Battery Powered Devices	NIC Version
aprel_rni.nic.image.prod.prod_sig.05.04.6043.03	5.4.6043	AP 5.0 Bridge 5.0 Relay 5.0 SocketAP 5.0		NIC 5xx
aprel_rni.nic_z.image.prod.prod_sig.05.04.6053.03	5.4.6053	MicroAP 5.0 (Zigbee)	IMU	NIC 5xx
slic_arm.nic.image.prod.prod_sig.05.04.0003.03	5.4.0003	Electricity Meter FSU 2.1	IMU 500S-INS	NIC 2xx
slic_arm.nic_z.image.prod.prod_sig.05.04.0013.03	5.4.0013	Electricity Meter (Zigbee)	IMU	NIC 2xx
slic_akio.nic.image.prod.prod_sig.05.04.0023.03	5.4.0023	Electricity Meter Street lights Load Control Switch FSU 4.0 FSU 4.5	IMU Milli 500S-INS	NIC 3xx/4xx
slic_akio.nic_z.image.prod.prod_sig.05.04.0033.03	5.4.0033	Electricity Meter (Zigbee)	IMU 500S-INS	NIC 3xx/4xx
slic_rni.nic.image.prod.prod_sig.05.04.0043.03	5.4.0043	Electricity Meter FSU 5.0	IMU Milli 500S-INS	NIC 5xx
slic_rni.nic_z.image.prod.prod_sig.05.04.0053.03	5.4.0053	Electricity Meter (Zigbee) Load Control Switch	IMU Milli 500S-INS	NIC 5xx

Resolved Issues

The following issue has been resolved in UtilOS 5.4.3.

3049533: Improper memory allocation issue

As a result of a memory buffer allocation management issue in UtilOS 5.4.0 that was triggered by certain conditions, devices could be prevented from establishing links or

receiving a Driver's License (DL) when attempting to establish a link. This issue is documented in *PIL-24-1606_UtilOS® 5.4:Memory Leak Issue.*

Resolution: This issue has been f ixed.

2904364: Correct settings for Extended Last Gasp (ELG) messaging for Australian Gen™5 devices supporting ELG

During the original branching for the UtilOS 5.x FW, updates impacted Australia's ELG requirements by incorrectly placing NIC's into an out of sync condition with the Australian Gen 5 devices' HW capability. This put NIC's into a hold on Last Gasp messages for the duration of the host meter's provided holdup time. NIC's were expected to continuously transmit messages until the holdup time was met.

Resolution: This issue has been f ixed.

Meter Compatibility

UtilOS firmware 5.4.3 has been tested with the meter firmware versions listed in the following table:

Table 2 Tested meter firmware versions

Meter	Meter Firmware (Gen 2,3,4)	Meter Firmware (Gen 5)
Aclara I-210+	1.0.13 2.2.1 2.3.1 3.8.1 4.0.8 4.2.1	
Aclara I-210+c	2.5.1 2.6.8 2.8.3 2.8.5 2.8.6 3.2.4 3.3.5 3.3.8 5.0.5	6.0.7
Aclara kV2c	4.11.8 4.12.2	4.12.2 4.14.7

Table 2 Tested meter firmware versions (continued)

Meter	Meter Firmware (Gen 2,3,4)	Meter Firmware (Gen 5)
Aclara kV2ce	4.0.10 4.10.5 4.11.8	4.12.2
EDMI Mk7A	1.47.4128 1.47.4149	
EDMI Mk7B	2.025 2.xxx	
EDMI Mk10D	1.47.16949	
EDMI Mk10E	2.027 2.071	2.33
Eletra Zeus	23Z.A1.F0.BR02.V0711	
Eletra Cronos 7023	FM34DHBLSMXC00101V57	
Honeywell DLMS AM660		01_03.00.00D-2f6f833
Honeywell A3	4.2.0 4.99.0 5.1.0 5.99.0	
Honeywell A4		02.05 02.06
Genus	G3627.125025 G3627.145002 GF637.325025 G3639.06 G3639.50	
Itron CENTRON ®	2.0.87 2.5.138 2.6.31 2.6.36 2.6.37 2.6.8	

Table 2 Tested meter firmware versions (continued)

Meter	Meter Firmware (Gen 2,3,4)	Meter Firmware (Gen 5)
Itron CENTRON II		2.7.45
Itron CENTRON Polyphase III		2.7.76
Itron EMx Single Phase DLMS RD meter		1.17
Itron SL7000	7.14 7.23	
Iskraemeco		ISK550TA06303038
Kamstrup	50981516-A55 50981516-B50 50981251-ZZ8	
L+G FOCUS AXe	5.33.512 5.63.778 5.64.6	5.70.90 5.70.99
L+ G FOCUS AX Polyphase	5.35.34 5.36.259 5.37.10 5.38.768	5.71.1025
L+G S4e	6.31 6.32	
L+G S4x		08.0C 08.0E
L+G U1200	5.5.R43	
L+G U1300	5.5.R43 5.5.R49	
L+G U3300	5.5.R43	
L+G U3400	5.5.R43	5.5.R49
Mirai	1210A5 FF.12.B4 M12BRT08	

Table 2 Tested meter firmware versions (continued)

Meter	Meter Firmware (Gen 2,3,4)	Meter Firmware (Gen 5)
Nansen NSX P 112i		S34S13 R3-02.BR.4604.V3.9.2
Nansen NSX P 213i		S34S13 R3-02.BR.4604.V3.9.2
Nansen Spectrum®	4.2 4.34 5.4	
Secure Premier 200	FPC7G10	
Secure Premier 210		FPGDG01
Secure Sprint 200	FPC6G20	
Secure Sprint 210		FPGCG01
Secure i-Credit 500	FPg1G01 FPh1G01 FPg1G1E FPh5106 FSB7G06 FSB3G06 FSB2G11	
Secure i-Credit 510		FPH7G01
Wasion		aM100-L-A0400128 aM300-L-C0400200 aM300-L-B0400114
XJ	90_0V	

Street Lights

UtilOS firmware 5.4.3 has been tested with the street light devices listed in the following table:

Table 3 Tested street light firmware versions

Streetlight Devices	Tested Firmware
Acuity	1.6.0 1.7.0
CIMCON Photocell	1.0.0019 1.0.0026
Rongwen	0.1.0012
SELC Photocell	01.00.01 01.00.05 02.00.02
Urban Control	2.0.0
OSRAM	0.0.6
SunRise	0.15

Firmware 5.4.2 Release Notes

Release Date: December 20, 2023

This release note describes new features and resolved issues in the UtilOS firmware 5.4.2 release.

Firmware Images

These images can be obtained from https://products.itron.com.

Table 4 UtilOS firmware 5.4.2 images

Image	5.4.2 Device Version	Supported Devices	Supported Battery Powered Devices	NIC Version
apr15_arm.nic.image.prod.prod_sig.05.04.5002.03	5.4.5002	AP 1.5 Relay 1.5		NIC 2xx
slic_arm.nic.image.prod.prod_sig.05.04.4002.03	5.4.4002	eBridge 2.0		NIC 2xx
slic_akio.nic.image.prod.prod_sig.05.04.4022.03	5.4.4022	Sentient Energy MM2/MM3 FCI		NIC 3xx/4xx
aprel_akio.nic.image.prod.prod_sig.05.04.6022.03	5.4.6022	AP 1.5i Relay 1.5i Bridge 4.0 AP 4.5 Relay 4.5 MicroAP™ for Streetlight cabinet with EDMI 10E		NIC 4xx
aprel_akio.nic_z.image.prod.prod_sig.05.04.6032.03	5.4.6032	MicroAP (Zigbee)	IMU 500S-INS	NIC 4xx

Table 4 UtilOS firmware 5.4.2 images (continued)

Image	5.4.2 Device Version	Supported Devices	Supported Battery Powered Devices	NIC Version
aprel_rni.nic.image.prod.prod_sig.05.04.6042.03	5.4.6042	AP 5.0 Bridge 5.0 Relay 5.0 SocketAP 5.0		NIC 5xx
aprel_rni.nic_z.image.prod.prod_sig.05.04.6052.03	5.4.6052	MicroAP 5.0 (Zigbee)	IMU	NIC 5xx
slic_arm.nic.image.prod.prod_sig.05.04.0002.03	5.4.0002	Electricity Meter FSU 2.1	IMU 500S-INS	NIC 2xx
slic_arm.nic_z.image.prod.prod_sig.05.04.0012.03	5.4.0012	Electricity Meter (Zigbee)	IMU	NIC 2xx
slic_akio.nic.image.prod.prod_sig.05.04.0022.03	5.4.0022	Electricity Meter Street lights Load Control Switch FSU 4.0 FSU 4.5	IMU Milli 500S-INS	NIC 3xx/4xx
slic_akio.nic_z.image.prod.prod_sig.05.04.0032.03	5.4.0032	Electricity Meter (Zigbee)	IMU 500S-INS	NIC 3xx/4xx
slic_rni.nic.image.prod.prod_sig.05.04.0042.03	5.4.0042	Electricity Meter FSU 5.0	IMU Milli 500S-INS	NIC 5xx
slic_rni.nic_z.image.prod.prod_sig.05.04.0052.03	5.4.0052	Electricity Meter (Zigbee) Load Control Switch	IMU Milli 500S-INS	NIC 5xx

New Features

This release contains the following new functionality:

Support for Gen™5 Secure meters

Firmware 5.4.2 supports Secure iCredit 510, Secure Spring 210, and Secure Premier 210 meters.

RED Regulatory updates for Europe

Firmware 5.4.2 supports updated Radio Equipment Directive (RED) regulatory requirements for Streetlight Cabinet Node and the Streetlight MicroAP®.

Resolved Issues

The following issues have been resolved in UtilOS 5.4.2.

2716146: Network congestion caused by improper management of AP Next Hop table

In deployments where non-Relay first hop devices could see several APs, there was a potential for network congestion leading to lower than expected Read success rate.

Because of a firmware error, APs did not manage their Next Hop table correctly when the configured maximum (nack_secondary_lvl) limit was reached. As a result, when an AP's Next Hop table was full, the AP would send a negative acknowledgment (NACK) response to a device's net register request (NREG) though the device was present in the table. This action forced the device to leave the AP and look for a new secondary AP. This behavior generated unnecessary control traffic leading to potential network congestion.

Resolution: Now, when the Next Hop table is full, the AP replies with a NACK only to devices that are not present in the table.

2980979: DA devices could reboot when a bundled message was received from the back office

Distribution Automation (DA) devices could reboot if there were 35 or more IP routes, and a bundled message from the back office including a next-hop command was received by the device.

Resolution: This issue has been fixed.

2981028: Meter node was unable to switch to a better AP

In deployments with very high AP utilization, a meter node was unable to register to an AP with a better cost if the secondary node limit (nack_second_lvl) was reached on this AP. Consequently, the meter node continued to be registered with the original AP even though the cost remained high.

Resolution: This issue has been fixed.

2981822: Gen™5 MicroAP™ configured with a MicroAP configuration file might not have shut down the Telit modem properly

In some cases, a Gen5 MicroAP device might not have shut down the Telit modem gracefully in the event of a power loss. This issue occurred if the WAN port number was not

set to the default value of **6** via the MicroAP configuration file. If this issue resulted in corrupting the modem, the modem had to be replaced.

Resolution: The configured WAN port number is stored and used to enable the Telit modem to shut down gracefully.

2982232: Running the `fwall show sas` command in GridScape resulted in infinite loop display for IPSec SA table

Upon receiving the `fwall show sas` command from the back office, a network infrastructure device in GridScape would send the same response in an infinite loop. Firmware entered this loop if the number of the secure association (SA) entries present on the node exceeded the size of the maximum response message.

Resolution: This issue has been fixed.

2982234: Default setting update for the AP ethernet monitoring functionality provided in 4.8.0 for faster recovery from DHCP-initiated IP changes for APs

UtilOS 4.6.0 introduced a feature that allowed for faster recovery from DHCP-initiated IP changes for APs. The feature, enabled by default, resets the AP if the cellular provider updated the IP address via DHCP. However, it was later identified that with some carriers, the proactive AP reset could result in the AP's NIC being out-of-sync with the cellular modem. A result of this potential out-of-sync condition was a loss of communication with the AP.

Resolution: The AP Ethernet monitoring feature is now disabled by default. Further information on this issue is found in *PIL-23-1584: Potential Loss of Communication to Access Point (AP) 5.0 with Some Carrier Networks.*

2982237/2884252: Master Bridges could fail to send RIP and MFP traffic when the head-end system (HES) router was rebooted

When the HES router was rebooted or if the Ethernet cable to a Master Bridge was disconnected and then reconnected, the device would stop sending Routing Information Protocol (RIP) and Master Failover Protocol (MFP) traffic updates to the HES router.

Resolution: This issue has been fixed.

2990980: The Zigbee/HAN radio was not identified when upgrading from UtilOS 3.x to 5.x on some Gen™2 NICs

Upgrades from UtilOS firmware 3.x to 5.x could cause the loss of the Zigbee configured parameters on some Gen 2 meter NICs because UtilOS 5.x did not properly identify the manufacturing settings for the Zigbee radio option and identified no Zigbee radio on the NIC.

Resolution: UtilOS firmware 5.4.2 corrects the loss of configuration when upgrading from a 3.x firmware version.

Meter Compatibility

UtilOS firmware 5.4.2 has been tested with the meter firmware versions listed in the following table:

Table 5 Tested meter firmware versions

Meter	Meter Firmware (Gen 2,3,4)	Meter Firmware (Gen 5)
Aclara I-210+	1.0.13 2.2.1 2.3.1 3.8.1 4.0.8 4.2.1	
Aclara I-210+c	2.5.1 2.6.8 2.8.3 2.8.5 2.8.6 3.2.4 3.3.5 3.3.8 5.0.5	6.0.7
Aclara kV2c	4.11.8 4.12.2	4.12.2 4.14.7
Aclara kV2ce	4.0.10 4.10.5 4.11.8	4.12.2
EDMI Mk7A	1.47.4128 1.47.4149	
EDMI Mk7B	2.025 2.xxx	
EDMI Mk10D	1.47.16949	
EDMI Mk10E	2.027 2.071	2.33
Eletra Zeus	23Z.A1.F0.BR02.V0711	

Table 5 Tested meter firmware versions (continued)

Meter	Meter Firmware (Gen 2,3,4)	Meter Firmware (Gen 5)
Eletra Cronos 7023	FM34DHBLSMXC00101V57	
Honeywell DLMS AM660		01_03.00.00D-2f6f833
Honeywell A3	4.2.0 4.99.0 5.1.0 5.99.0	
Honeywell A4		02.05 02.06
Genus	G3627.125025 G3627.145002 GF637.325025 G3639.06 G3639.50	
Itron CENTRON ®	2.0.87 2.5.138 2.6.31 2.6.36 2.6.37 2.6.8	
Itron CENTRON II		2.7.45
Itron CENTRON Polyphase III		2.7.76
Itron EMx Single Phase DLMS RD meter		1.17
Itron SL7000	7.14 7.23	
Iskraemeco		ISK550TA06303038
Kamstrup	50981516-A55 50981516-B50 50981251-ZZ8	

Table 5 Tested meter firmware versions (continued)

Meter	Meter Firmware (Gen 2,3,4)	Meter Firmware (Gen 5)
L+G FOCUS AXe	5.33.512 5.63.778 5.64.6	5.70.90 5.70.99
L+ G FOCUS AX Polyphase	5.35.34 5.36.259 5.37.10 5.38.768	5.71.1025
L+G S4e	6.31 6.32	
L+G S4x		08.0C 08.0E
L+G U1200	5.5.R43	
L+G U1300	5.5.R43 5.5.R49	
L+G U3300	5.5.R43	
L+G U3400	5.5.R43	5.5.R49
Mirai	1210A5 FF.12.B4 M12BRT08	
Nansen NSX P 112i		S34S13 R3-02.BR.4604.V3.9.2
Nansen NSX P 213i		S34S13 R3-02.BR.4604.V3.9.2
Nansen Spectrum®	4.2 4.34 5.4	
Secure Premier 200	FPC7G10	
Secure Premier 210		FPGDG01
Secure Sprint 200	FPC6G20	

Table 5 Tested meter firmware versions (continued)

Meter	Meter Firmware (Gen 2,3,4)	Meter Firmware (Gen 5)
Secure Sprint 210		FPGCG01
Secure i-Credit 500	FPg1G01 FPh1G01 FPg1G1E FPh5106 FSB7G06 FSB3G06 FSB2G11	
Secure i-Credit 510		FPH7G01
Wasion		aM100-L-A0400128 aM300-L-C0400200 aM300-L-B0400114
XJ	90_0V	

Note: Upgrading a Load Control Switch from a 3.x version to a 5.x version will require assistance from your Itron customer representative team.

Street Lights

UtilOS firmware 5.42 has been tested with the street light devices listed in the following table:

Table 6 Tested street light firmware versions

Streetlight Devices	Tested Firmware
Acuity	1.6.0 1.7.0
CIMCON Photocell	1.0.0019 1.0.0026
Rongwen	0.1.0012
SELC Photocell	01.00.01 01.00.05 02.00.02

Table 6 Tested street light firmware versions (continued)

Streetlight Devices	Tested Firmware
Urban Control	2.0.0
OSRAM	0.0.6
SunRise	0.15

Firmware 5.4.1 Release Notes

Firmware 5.4.1 was an internal release.

Firmware 5.4.0 Release Notes

Release Date: April 15, 2022

This release note document describes the enhancements, resolved issues, and known issues in the UtilOS firmware 5.4.0 release.

Firmware Images

These images can be obtained from https://products.itron.com.

Table 7 UtilOS firmware 5.4.0 images

Image	5.4.0 Device Version	Supported Devices	Supported Battery Powered Devices	NIC Version
apr15_arm.nic.image.prod.prod_sig.05.04.5000.03	5.4.5000	AP 1.5 Relay 1.5		NIC 2xx
slic_arm.nic.image.prod.prod_sig.05.04.4000.03	5.4.4000	eBridge 2.0		NIC 2xx
slic_akio.nic.image.prod.prod_sig.05.04.4020.03	5.4.4020	Sentient Energy MM2/MM3 FCI		NIC 3xx/4xx
aprel_akio.nic.image.prod.prod_sig.05.04.6020.03	5.4.6020	AP 1.5i Relay 1.5i Bridge 4.0 AP 4.5 Relay 4.5 MicroAP™ for Streetlight cabinet with EDMI 10E		NIC 4xx
aprel_akio.nic_z.image.prod.prod_sig.05.04.6030.03	5.4.6030	MicroAP (Zigbee)	IMU 500S-INS	NIC 4xx

Table 7 UtilOS firmware 5.4.0 images (continued)

Image	5.4.0 Device Version	Supported Devices	Supported Battery Powered Devices	NIC Version
aprel_rni.nic.image.prod.prod_sig.05.04.6040.03	5.4.6040	AP 5.0 Bridge 5.0 Relay 5.0 SocketAP 5.0		NIC 5xx
aprel_rni.nic_z.image.prod.prod_sig.05.04.6050.03	5.4.6050	MicroAP 5.0 (Zigbee)	IMU	NIC 5xx
slic_arm.nic.image.prod.prod_sig.05.04.0000.03	5.4.0000	Electricity Meter FSU 2.1	IMU 500S-INS	NIC 2xx
slic_arm.nic_z.image.prod.prod_sig.05.04.0010.03	5.4.0010	Electricity Meter (Zigbee)	IMU	NIC 2xx
slic_akio.nic.image.prod.prod_sig.05.04.0020.03	5.4.0020	Electricity Meter Street lights Load Control Switch FSU 4.0 FSU 4.5	IMU Milli 500S-INS	NIC 3xx/4xx
slic_akio.nic_z.image.prod.prod_sig.05.04.0030.03	5.4.0030	Electricity Meter (Zigbee)	IMU 500S-INS	NIC 3xx/4xx
slic_rni.nic.image.prod.prod_sig.05.04.0040.03	5.4.0040	Electricity Meter FSU 5.0	IMU Milli 500S-INS	NIC 5xx
slic_rni.nic_z.image.prod.prod_sig.05.04.0050.03	5.4.0050	Electricity Meter (Zigbee) Load Control Switch	IMU Milli 500S-INS	NIC 5xx

New Features and Enhancements

This release contains the following new or updated functionality:

Support for Secure Meter's new switch mode

Support has been added for Secure Meter's new auto-disconnect switch mode.

OFDM offset tables support maximum RF output power limit lower than 30dBm

To ensure compliance of Orthogonal Frequency Division Multiplexing (OFDM) power for regional limits specifications where the RF output power limit is lower than 30 dBm, the OFDM offset tables have been adjusted to limit the maximum output power.

Default EIRP power output for Thailand reduced

To satisfy Thailand's (CC764) regulatory requirements, the EIRP power output has been reduced to a maximum of 27dBm.

Support for EDMI Mk7BT meters

Support has been added for EDMI Singlephase Mk7BT meters.

Support for Zigbee simple metering cluster

Support has been added for Zigbee SEP 1.1 simple metering clusters for Honeywell A4 meters.

Support for new country codes

In addition to the country codes supported in previous UtilOS firmware releases, support has been added for the following country codes:

- EUB10: DEU, ITA, NLD, TUR
- EUB 14: CYP, FRA, GRC, LVA, LTU, POL, ROU

Support for improved overhead in data retrieved from SensorIQ profile

The high overhead in NIC response messages to SensorIQ for high-frequency reads resulted in an increase in network traffic due to multiple packets. Now, the overhead of the response message format (protobuf) has been lowered to reduce network traffic.

Updated Treck TCP/IP stack

The Treck TCP/IP stack has been updated to allow UtilOS 5.4.0 and future releases to rapidly implement protection against any security vulnerabilities reported by Treck.

Support for 500S Riva modules in GenX networks

Support is provided for Itron Riva 500G (gas) and 500W (water) modules to transition from a Riva network to a GenX network.

Support for Large Failover Groups in an AP-based DA network

To enable customers to scale their IPv4 AP-based DA (Distribution Automation) networks, support has been added for large failover groups (LFOGs). An LFOG group contains 500 or

more devices. To support LFOG, each network will need to use a separate, unique MFP (Master Failover Protocol) address. By default, this feature is enabled.

> **Note:** When LFOG is enabled, IPARM on IPv6 is automatically disabled. As a result, Bridges are unable to route data through meters. Running the `iparm_update_v6_route 1` netmgr command will enable IPARM but will disable LFOG. To enable routing through meters, contact your Itron delivery team.

Support for new water meter

Support is provided for Diehl water meters.

1860658: Enhanced support for Itron Edge Gateway Router

Support for IoTR battery-powered devices (BPDs) in the Itron Mesh Network has been enhanced as follows:

- Support for the use of Driver's License Certificate Authority (DLCA) for link-layer security.
- Support for Light Emitting Diode (LED) to communicate network status.

Support for Gen5 Nansen meters

Support has been added for Nansen NSX P 112i and 213i meters.

Resolved Issues

This section lists issues that have been addressed in the 5.4.0.

2380226: Delayed Extended Last Gasp (ELG) messages

During power outages, a bundled ELG message sent to the AP via neighboring nodes could be delayed for hours or not received at all. This issue occurred due to one or both of the following reasons:

- Inability of a standard Last Gasp (LG) device running UtilOS firmware 3.x to forward ELG message bundles. This resulted in dropped packets.
- The ELG message bundle was stuck in a loop. This occurred when a customer did not reboot the devices after editing their configuration to change the destination for ELG messages.

Resolution: To enable standard LG devices to forward ELG messages, the customer must upgrade devices running firmware 3.*x* to firmware 5.4.0. To prevent loops, UtilOS 5.4.0 performs configuration changes dynamically, and hence, devices do not need a reboot.

2150793: UtilOS firmware did not recognize error response codes from Load Control Switch (LCS) devices

UtilOS firmware could only recognize an "OK" (0x00) response code from LCS devices. Hence, if an LCS device rejected a request for new keys for any reason, the firmware did not recognize the rejection code. It discarded the valid keys and updated its sysvar with invalid keys. This resulted in a communication failure.

Resolution: The firmware now recognizes all the response codes generated by an LCS device for Key Update requests. During the Key Update process, if the NIC reboots or if there are errors, such as Flash read/write errors, the NIC attempts to recover the key.

2565734: Battery calibration could restart frequently

Gen4 and Gen5 APs and Relays configured with Battery Backup Units (BBUs) and UtilOS® firmware 5.0, 5.0.1, 5.2.0 or 5.2.1 could encounter an issue in which the battery management system's calibration cycle ran at a greater frequency than the expected 62 days. In some cases, a calibration cycle could begin within 24 hours after the completion of the previous calibration cycle.

Resolution: This issue has been fixed.

2612723: Updated first capture binding process may incorrectly allow for a new operator certificate to form a secure association

The NIC, after expired certificates were cleared, incorrectly allowed a previously bound device to form a secure association using a valid certificate with a different `subjectDN`. An update for enhanced first capture functionality in the UtilOS 5.2.0 release did not ensure that a device bound, with an operator certificate that has since expired, maintain its binding to the existing operator and refuse new valid operator certificates with a differing operator identified by the `subjectDN`. The behavior mirrored the activity experienced in previous versions of firmware in which the NIC upon power up after clearing the certificates could join a valid GenX network if its network ID matched.

Resolution: The firmware now identifies if its previously bound operator does not match a new valid certificates operator, it will reject the new certificate.

Meter Compatibility

UtilOS firmware 5.4.0 has been tested with the meter firmware versions listed in the following table:

Table 8 Tested meter firmware versions

Meter	Meter Firmware (Gen 2,3,4)	Meter Firmware (Gen 5)
Aclara I-210+	1.0.13 2.2.1 2.3.1 3.8.1 4.0.8 4.2.1	
Aclara I-210+c	2.5.1 2.6.8 2.8.3 2.8.5 2.8.6 3.2.4 3.3.5 3.3.8 5.0.5	5.1.0 6.0.7
Aclara kV2c	4.11.8 4.12.2	4.12.2 4.14.7
Aclara kV2ce	4.0.10 4.10.5 4.11.8	4.12.2
EDMI Mk7A	1.47.4128 1.47.4149	
EDMI Mk7B	2.025 2.xxx	
EDMI Mk10D	1.47.16949	
EDMI Mk10E	2.027 2.028 2.071	2.33
Eletra Zeus	23Z.A1.F0.BR02.V0711	
Eletra Cronos 7023	FM34DHBLSMXC00101V57 FW00374V22	

Table 8 Tested meter firmware versions (continued)

Meter	Meter Firmware (Gen 2,3,4)	Meter Firmware (Gen 5)
Honeywell DLMS AM660		01_02.00.00-9cd5c56[M] 01_03.00.00D-2f6f833
Honeywell A3	4.2.0 4.99.0 5.1.0 5.99.0	
Honeywell A4		02.05 02.06
Genus	G3627.125025 G3627.145002 GF637.325025 G3639.06 G3639.50	
Itron CENTRON ®	2.0.87 2.5.138 2.6.31 2.6.36 2.6.37 2.6.8	
Itron CENTRON II		2.7.45
Itron CENTRON Polyphase III		2.7.74
Itron EMx Single Phase DLMS RD meter		1.17
Itron SL7000	7.14 7.23	
Iskraemeco		ISK550TA06303038
Kamstrup	50981516-A55 50981516-B50 50981251-ZZ8	

Table 8 Tested meter firmware versions (continued)

Meter	Meter Firmware (Gen 2,3,4)	Meter Firmware (Gen 5)
L+G FOCUS AXe	5.33.512 5.63.778 5.64.6	5.70.90 5.70.99
L+ G FOCUS AX Polyphase	5.35.34 5.36.259 5.37.10 5.38.768 5.39.1046	5.71.1025
L+G S4e	6.31 6.32	
L+G S4x		08.0C 08.0E
L+G U1200	5.5.R43	
L+G U1300	5.5.R43 5.5.R49	
L+G U3300	5.5.R43	
L+G U3400	5.5.R43	5.5.R49
Mirai	1210A5 FF.12.B4 M12BRT08	
Nansen NSX P 112i		S34S13 R3-02.BR.4604.V3.9.2
Nansen NSX P 213i		S34S13 R3-02.BR.4604.V3.9.2
Nansen Spectrum®	4.2 4.34 5.4	
Secure Premier 200	FPC7G10	
Secure Sprint 200	FPC6G20	

Table 8 Tested meter firmware versions (continued)

Meter	Meter Firmware (Gen 2,3,4)	Meter Firmware (Gen 5)
Secure i-Credit 500	FPg1G01 FPh1G01 FPg1G1E FPh5106 FSB7G06 FSB3G06 FSB2G11	
Wasion		aM100-L-A0400128 aM300-L-C0400200 aM300-L-B0400114
XJ	90_0V	

Note: Upgrading a Load Control Switch from a 3.x version to a 5.x version will require assistance from your Itron customer representative team.

Streetlights

UtilOS firmware 5.4.0 has been tested with the streetlight devices listed in the following table:

Table 9 Tested street light firmware versions

Streetlight Devices	Tested Firmware
Acuity	1.6.0 1.7.0
CIMCON Photocell	1.0.0014 1.0.0019 1.0.0026
enLight®	1.0.0.Scd220 Dali 1.0.1 Scd220
Rongwen	0.0.0008 0.1.0012

Table 9 Tested street light firmware versions (continued)

Streetlight Devices	Tested Firmware
SELC Photocell	01.00.01 01.00.05 01.00.07 02.00.02
Urban Control	2.0.0
OSRAM	0.0.6
SunRise	0.15

Firmware Versioning and Naming Conventions

This section explains NIC firmware versioning in 5.0.0 and later releases, and provides details on the naming conventions used for firmware images.

Convention

Firmware adheres to the following versioning scheme:

Figure 1 Firmware Versioning Convention

The third segment of numbers has four positions, the first of which specifies the device type, as shown in the table below:

Table 10 Device type indicator

Code	Meaning
0	Electric meter
4	eBridge 2.0; Sentient FCI
5	AP and Relay 1.5
6	Access Point / Relay 1.5i, 4.5, 5.0; MicroAP, MicroAP 5.0; SocketAP 5.0, Bridge 4.0, 5.0.

In the third segment, the second position is not currently used externally. The third position represents the NIC Generation and if the version is Zigbee-capable. The fourth position represents the service pack.

Refer to the table below for the numerical representation of the NIC Generation and Zigbee availability.

Table 11 NIC generation and Zigbee availability

Code	NIC/Zigbee
0	Gen2
1	Gen2 + Zigbee

Table 11 NIC generation and Zigbee availability (continued)

Code	NIC/Zigbee
2	Gen3 and Gen4
3	Gen3 and Gen4 + Zigbee
4	Gen5
5	Gen5 + Zigbee

Image Naming Convention

Image names adhere to the conventions shown below.

> **Note:** Builds with HAN enabled also include `nic_z` (electricity meters only) as part of the image name.

deviceType_nicProcessor.nicType.signedBuild.imageVersion.securityVersion

where:

- *deviceType* is one of the following:
  - apr15 (AP/Relay 1.5)
  - slic (electric meter NIC or eBridge 2.0)
  - aprel (AP/Relay/Bridge/MicroAP/SocketAP)
- *nicProcessor* is one of the following:
  - arm: NIC 2xx (Gen2)
  - akio: NIC 3*xx*/NIC 4*xx* (Gen3/Gen4)
  - rni: NIC 5*xx* (Gen5)
- *nicType* is one of the following:
  - nic (non-Zigbee (HAN) version)
  - nic_z (Zigbee version)
- *signedBuild* is:
  - prod_sig (production signed)
- *imageVersion* is one of the image versions listed in Firmware Versioning and Naming Conventions on page 32.
- *securityVersion* is the last two digits of the image version number.

Example:

slic_rni.nic_z.image.prod.prod_sig.05.02.0050.03

where:

- slic = Electric meter
- rni =NIC processor (NIC 5*xx*)

- nic_z = Zigbee capable
- prod.prod_sig= production signed
- Image version =05.02.5000 (5.2.0)
- Security verison = 03